

Riccardo Barone

07796416717 | riccardo.work.ul@gmail.com | [LinkedIn](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

A professional and highly motivated technologist with a versatile academic background in **Computer Science**. I bring a mature, collaborative approach to problem-solving and a proven work ethic developed through managing high-pressure environments.

I am a strategic thinker with a strong focus on the ethical application of technology. Known for my adaptability and clear communication, I excel at translating complex technical concepts into actionable insights. I am eager to apply my analytical mindset and professional reliability to help drive secure, customer-focused innovation within a forward-thinking team.

PROJECTS

Full-Stack Modular System Design: Stock Management |

Focus: Software Architecture & Development | **Tech:** Java, Hibernate (ORM), MySQL, Aiven Cloud

- Engineered a high-availability stock management system using a modular 3-Tier Architecture (DTO, DAO, and Service Layers) to ensure strict separation of concerns and code maintainability.
- Implemented Hibernate ORM for efficient data persistence and object-relational mapping, optimizing database interactions for a cloud-hosted MySQL environment on Aiven.
- Applied Core Object-Oriented Programming (OOP) principles to create a scalable and robust codebase, focusing on design patterns that support enterprise-level software requirements

Secure Enterprise Network: High-Availability Infrastructure | [GitHub](#)

Focus: Network Engineering & Security | **Tech:** Cisco Packet Tracer, GLBP, EIGRP, LACP, VPNs, AAA, ACLs

- Modernized a legacy infrastructure into a secure 3-Tier Topology, implementing routing protocols at the for inter-VLAN routing with active load balancing and redundant failover mechanisms.
- Hardened the network against unauthorized access and Layer 2 attacks by implementing VLAN Segmentation, Port Security, and Access Control Lists (ACLs) to mitigate threats.
- Established secure, encrypted communication between distributed sites using Site-to-Site VPNs and implemented AAA to centralize and audit administrative access.

Offensive Security: System Auditing & Penetration Testing |

Focus: Cyber Security & Ethical Hacking | **Tech:** Metasploit, Nmap, OpenVAS, MSFvenom, PowerUp

- Executed a full-cycle penetration test following the OSI Model and CIA Triad principles to identify critical RCE vulnerabilities and map a target's attack surface.
- Leveraged Metasploit and MSFvenom to exploit system flaws and performed advanced Privilege Escalation by hijacking misconfigured services to gain SYSTEM-level access.
- Conducted post-exploitation auditing including credential harvesting via hashdump and NTLM password cracking attempts using Hashcat to assess organizational password policy strength.

Data Science & Applied Statistical Intelligence |

Focus: Knowledge Discovery & Hypothesis Testing | **Tech:** Weka, Python (Pandas), SQL

- Managed the end-to-end data lifecycle using CRISP-DM and KDD to translate complex statistical findings into actionable strategy and commercial value.
- Executed a range of inferential statistics to validate hypotheses, including Independent T-Tests for group comparisons, Correlation analysis, and Mediation Analysis to identify underlying causal mechanisms.
- Leveraged Weka to perform Market Basket Analysis, interpreting Support, Confidence, and Lift measures to identify and quantify statistically significant associations within large-scale datasets.

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TECHNICAL SKILLS

Programming and Architecture | Python, Java, JavaScript, TypeScript, React, OOP, UML Modelling, MVC architecture, Design principles, Design Patterns.

Data Engineering | Database Architecture principles (normalisation), Relational databases (SQL, Hibernate), NO-SQL Databases (MongoDB, Mongoose).

Data Analytics | Crisp-DM, KDD, Statistical Analysis, Market Basket Analysis, business insight generation

Network Engineering | Layer 2/3 Architecture, 3-tier & 2-Tier topologies, Routing protocols (BGP, EIGRP, GLBP, LACP).

Network Security | VPN, Port security, Access Control Lists (ACL), AAA Authentication, VLAN Segmentation.

Cyber Security & pen testing | Penetration Testing Lifecycle, Nmap, OpenVAS, Burp Suite, Metasploit, Password Cracking (Hashcat, John the Ripper).

EDUCATION

BSc (Hons) Computer Science | University of Winchester | Expected Graduation: 2026

Key modules: AI and machine learning (Python), Database analysis and design (SQL), Cloud Computing infrastructure, Distributed systems, insight Management, Penetration testing

Professional Diploma in Culinary Arts & Kitchen Management (UK Level 3 Equivalent) | Italy | 2020

Focus on professional standards, operational management, and leadership in high-pressure environments.

WORK EXPERIENCE

Freelance Web Development | [Winchester Eagles volleyball Club](#) | July 2025 – Sept 2025

- Managed the end-to-end project lifecycle, translating non-technical stakeholder requirements into a functional, modern web presence.
- Architected and deployed a responsive application using Next.js, TypeScript, and React, managed through a Vercel deployment pipeline for high availability.
- Provided ongoing technical mentorship to club members, ensuring a smooth transition to the new system and high stakeholder satisfaction.

Second Chef | New Forest nursing Home | Jan 2024 – Present

- Act as the primary lead for kitchen operations during senior management absence, ensuring 100% compliance with strict healthcare and safety regulations.
- Resolve real-time logistical challenges for multidisciplinary teams and adapt services to meet complex, high-priority resident requirements.
- Oversee inventory and supply chain coordination, maintaining budget efficiency through methodical waste reduction and cost control.

Junior Sous-Chef | [Balmer Lawn Hotel] | April 2022 – Jan 2024

- Coordinated with senior management and front-of-house teams to maintain high-performance standards during peak service hours.
- Mentored and trained junior staff and apprentices, fostering a collaborative culture and ensuring high standards of technical proficiency.
- Facilitated clear communication across multidisciplinary kitchen stations to optimize workflow and resolve operational bottlenecks in real-time.